
The Capacity Lever Changes Sign: Boundary Optima and Why Interior Sweeps Mislead

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Abstract

We report a lever whose sign depends on the operating point, and a diagnosis of why twenty-two directions in this campaign returned nothing. Sorting every closed direction by its effect on tokens consumed collapses the campaign into four rows: interventions that do not move tokens land within $\pm 1.3\text{e-}3$ against a $2.6\text{e-}3$ gate, while interventions that move tokens either are already captured by torch.compile or push training past a repetition wall worth 16–18 gates. The frame therefore has essentially one free parameter, and the best configuration already sits just under the wall at 1.9 passes over a frozen corpus. This is a boundary optimum, which explains the null results directly: the gradient is zero in the feasible directions and a cliff in the one that matters. The wall itself is a property of tokens and not of model shape — three configurations spanning 39.8M–94.4M dense parameters and three different shapes cross it at 2.3–2.7 passes — and shuffling data order moves it by only 4.7% of its size, confirming repetition rather than corpus region as the cause. Reviewing the campaign grid then exposed an empty cell: n-gram capacity had been varied only below the wall and regularisation only above it. Filling it, cutting hash-table rows by $4\times$ improves the over-the-wall configuration by 0.0163 (6.2 gates), where an earlier sweep measured a comparable cut costing +0.0079 below the wall. The sign of the capacity lever is a property of the operating point, not of the lever.

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1. Twenty-Two Directions, Four Rows

Sorted by what each lever does to tokens consumed:

lever class	tokens	outcome
intrinsic refinements	none	all $\pm 1.3\text{e-}3$
compile-captured	none	8 killed prospectively
depth 8→6	+62%	−0.002002, 0.77 gates
depth 5 / dim 512 / $2\times$	+100–140%	collapse, 16–18 gates

The adoption gate is 0.002614; the intrinsic channel tops out near 0.0013; the throughput channel is bounded above by a cliff. The window between “too small to matter” and “over the edge” is narrow, and the best measured configuration — depth 6, dim 640, 483M tokens — sits inside it. At a boundary optimum, local search reports zero gradient in every direction it is allowed to move and a cliff in the one it is not. That is the campaign’s null-result pattern, and it is a property of the frame rather than of the interventions.

2. The Wall Is About Tokens, Not Shape

configuration	steps	tokens	passes	val_bpb
depth 6, dim 640	3276	483M	1.9	0.9296
$2\times$ budget, d8	3918	578M	2.3	0.9870
depth 6, dim 512	4228	623M	2.5	0.9729
depth 5, dim 512	4528	668M	2.7	0.9784

Three shapes, 39.8M–94.4M dense parameters, one boundary. Separately, shuffling row-group order at depth 5 moves the endpoint by 0.0022 — 4.7% of the 0.0464 cliff — against preregistered thresholds of <0.003 for a repetition cause and >0.01 for a corpus-region cause. The cause is repetition. This mattered because shard order is fixed, so “the third pass” and “the tail of the corpus for the third time” had been perfectly confounded in every run the campaign had ever done.

3. What Regularisation Does and Does Not Do

Two attempts failed before the right one. Global weight decay $\{0.1, 0.3, 0.6\}$ degraded monotonically; reading the optimizer groups showed why — it reaches only the dense matrices, 3.2% of parameters, while the 2.82e9-parameter hash tables (96.8% of the model) use a separate coefficient defaulting to zero. Correcting that, decay applied to the tables closed the train/val gap by 93% ($+0.2795 \rightarrow +0.0201$) and the endpoint still worsened at both depths. In a hashed n -gram memory the memorising rows and the signal-carrying rows are the same rows, and a norm penalty is indiscriminate across rows.

4. The Empty Cell, and the Sign Flip

	below wall (1.9)	above wall (2.7)
capacity	+0.0079, +0.0109	never tested
regularisation	+0.0349	+0.0128

Capacity had been varied only where there is no overfitting to remove, and regularisation only where there is. The cell was empty because a prior result was reused outside the operating point that produced it — “capacity was already swept” was true and irrelevant.

Filling it, at depth 5 (2.7 passes), $n=4$, 12/12 arms clean:

mult	val_bpb	train	val	gap
64	0.979248	2.4838	2.7546	+0.2708
16	0.962948	2.6034	2.6954	+0.0920
4	0.969048	2.6786	2.6783	-0.0003

-0.016300 at mult 16, 6.2 gates better, against a pre-registered threshold of 0.01. The earlier sweep measured a comparable cut costing $+0.0079$ below the wall. The sign flips.

And it is not decay in disguise. Decay closed the gap while losing the endpoint; cutting rows moves the gap $+0.2708 \rightarrow +0.0920$ and improves the endpoint, with training loss rising only $2.4838 \rightarrow 2.6034$ while validation falls $2.7546 \rightarrow 2.6954$. Shrinking every row removes signal and memorisation together. Removing rows forces distinct contexts to collide into a shared row, which is a structural constraint toward generalisation. Same phrase, different operator.

Scope. This recovers 33% of the cliff, and the rescued configuration (0.962948) remains 0.0334 worse than the best one (0.929570). It is a mechanism result, not an adoption.

5. Next Experiments

5.1. E22 — Does the sign flip reach the best configuration?

(running)

- Claim. Internal, from E8 and E21 above. No external claim.
- Mechanism. Observed: the capacity lever costs $+0.0079$ below the wall and gains 0.0163 above it. Proposed cause: row count sets memorisation capacity; below threshold rows carry signal and removing them costs, above it surplus rows store memorised continuations and removing them pays. Rival: the flip tracks the train/val gap, not the pass count — depth 6’s gap is -0.0663 (validation better than training), so cuts must hurt there. Separator: the sign at depth 6, the one point where pass count is near the boundary but the gap is negative.
- Hypothesis. depth $6 \times \text{mult } \{64, 32, 16\}$, $n=4$. Anything beating mult 64 by ≥ 1 gate goes to $n=10$ through the full funnel before any claim.
- Reasoning. Against, and this is my honest prior: E8 measured the cut costing at a comparable pass count, and depth 6 is not overfitting by the gap measure, so harm is expected. For: 1.9 passes is close to the flip and the flip location is unmeasured. Cost is nil — E21 measured step time unchanged by mult (66.5 vs 67.0 ms).
- Ratings. 3 / 4 / 5 / 5 / 4 / 5 / 5.

5.2. E23 — Locate the flip

- Claim. Internal.
- Mechanism. Observed: sign is negative at 1.2–2.0 passes and positive at 2.7. Cause: a threshold in tokens-per-row. Rival: a threshold in the train/val gap. Separator: these predict different flip locations when pass count and gap are varied independently, which TIME_BUDGET and depth do separately.
- Hypothesis. Sweep TIME_BUDGET $\in \{300, 375, 450\}$ at fixed depth $6 \times \text{mult } \{64, 16\}$, $n=3$, and locate the crossing. Decision rule: if the crossing tracks passes it should sit near 2.2–2.4 regardless of depth.
- Reasoning. For: a located threshold turns an observed sign flip into a predictive rule, and the campaign has been repeatedly burned by extrapolating levers across operating points. Against:

it changes the charged budget, so results are not comparable to the frame's gate and this is measurement, not a candidate.

- Ratings. 2 / 5 / 5 / 3 / 5 / 4 / 5.

6. Conclusion

A campaign of null results is not always a campaign of bad ideas. Here it was a boundary optimum: one free parameter, a baseline already at its edge, and a cliff in the only direction with headroom. The one lever that moved the cliff was found not by a new idea but by noticing that a cell of the campaign's own grid was empty — capacity had been swept only where the phenomenon it acts on does not exist. The general lesson is that a lever's sign can belong to the operating point rather than to the lever, and that a result which rules something out is scoped to the regime that produced it.